

Instructions

There are **50 total points**. This exam is closed-book and closed-notes except for **one 8.5 × 11-inch sheet of paper with handwritten notes (both sides permitted)**. No electronic devices, additional pages, or other materials are permitted. When asked to provide your answer within a figure or table, be careful to not exceed box boundaries. Bubbles must be filled out completely: ● is correct, ✓ ⊙ ✗ are incorrect. **1.** Write your full name in the box to acknowledge the instructions.

Motivating Applications

1. [2 points]: Which of the following are examples of application quality metrics that machine learning models can help predict in video streaming systems?

Select all that apply.

- | | |
|--|---|
| ☐ Video resolution (e.g., 480p, 720p, 1080p) | ☐ Rebuffering events (video stalling) |
| ☐ Startup delay before playback begins | ☐ WiFi signal strength at the access point |
| ☐ Hard disk read/write speed | ☐ Number of quality changes during playback |

2. [3 points]: In adaptive bitrate (ABR) video streaming, a player observes that network bandwidth has temporarily increased. Would the player always immediately request the highest available quality? ☐ Yes ☐ No

Explain why or why not.

(Answer inside the box)

3. [3 points]: Which of the following network features would be most useful for distinguishing between normal HTTP web browsing and HTTP scanning activity (e.g., Log4j vulnerability scanning)? Select all that apply.

- | | |
|---|---|
| ☐ Average packet size | ☐ Number of unique destination IPs per source |
| ☐ Rate of connection attempts (connections/sec) | ☐ Source IP geographic location |
| ☐ Ratio of failed to successful connections | ☐ Number of unique User-Agent strings |

4. [3 points]: Which of the following traffic features would be most useful for detecting automated web scraping (e.g., downloading all posts from a social media profile) versus normal human browsing? Select all that apply.

- | | |
|---|---|
| ☐ Rate of page requests (requests/second) | ☐ Source IP address |
| ☐ Time of day of requests | ☐ Sequential access to pages from same author |
| ☐ Time between requests vs. human reading speed | ☐ Destination TCP port (80 or 443) |

5. [4 points]: DNS-based features can detect malware even when application traffic is encrypted (HTTPS). ☐ Yes ☐ No

Explain why or why not.

(Answer inside the box)

Initials: _____

Data Acquisition

6. [3 points]: Which of the following are characteristics of passive network measurement? Select all that apply.

- | | |
|---|---|
| ☐ Generates test packets to measure performance | ☐ Observes real user traffic and behavior |
| ☐ Does not disrupt network conditions | ☐ Can be performed on-demand at any time |
| ☐ Requires no special network access | ☐ May raise privacy concerns (GDPR, etc.) |

7. [2 points]: A network flow is defined by which 5 header fields? Select all that apply.

- | | |
|--|---|
| ☐ Source IP address | ☐ Packet timestamp |
| ☐ Destination IP address | ☐ Source port |
| ☐ Destination port | ☐ Protocol (TCP, UDP, etc.) |

8. [3 points]: What is one advantage and one disadvantage of using flow records compared to full packet captures for machine learning applications?

(Answer inside the box)

ML Pipeline and Data Quality

9. [4 points]: In Assignment 1, you identified video segments in Netflix traffic by detecting gaps (e.g., 1-second pauses) between packets rather than using fixed byte counts.

- (a) Why are video segments variable in size rather than always the same number of bytes?
- (b) Name one feature you could compute from segment-level statistics that would be useful for predicting video resolution.

(Answer inside the box)

10. [3 points]: A model trained on university campus network traffic might fail when deployed on an enterprise network because the training data is non-representative.

☐ Yes ☐ No

Why or why not?

(Answer inside the box)

Initials: _____

11. [3 points]: In class, we discussed the TTL (Time-To-Live) example from network attack detection. A model achieved high accuracy but relied heavily on the TTL field.

(a) Briefly explain what TTL represents and why it's problematic for attack detection.

(b) Suggest one strategy to prevent over-relying on this feature.

(Answer inside the box)

Model Training and Evaluation

12. [3 points]: At optimal model complexity, test error is minimized in the bias-variance tradeoff.

☐ Yes ☐ No

What happens to model accuracy on the test data at high complexity?

(Answer inside the box)

13. [3 points]: When preparing features for your model, you normalize throughput across all data, and after that split 80/20 into train/test sets. Is this a correct data preparation approach?

☐ Yes ☐ No

Why or why not?

(Answer inside the box)

14. [2 points]: Which metrics are better than accuracy for evaluating spam detection (90% spam rate)? Select all that apply.

☐ Precision

☐ Total number of emails processed

☐ Recall / TPR (of actual spam, what fraction caught)

☐ F1 Score (harmonic mean of precision and recall)

☐ Confusion matrix showing all classifications

☐ False Positive Rate (legitimate emails marked spam)

Initials: _____

Cross-Validation and Model Drift

15. [3 points]: K-fold cross-validation provides a more robust estimate of model generalization than a single train-validation split. ☐ Yes ☐ No

Why or why not?

(Answer inside the box)

16. [4 points]: During COVID-19, campus traffic patterns changed dramatically. A pre-COVID trained model failed during COVID.

(a) What is this phenomenon called?

(b) Why doesn't cross-validation protect against this?

(Answer inside the box)

Feedback

17. [2 points]: Interest (1=Boring!; 10=Amazing!):

Difficulty (1=Too easy; 10=Too hard):

One thing you like. One suggestion for improvement:

(Answer inside the box)